

Linear Search

Input: A sequence of n numbers

$A = \langle a_1, a_2, \dots, a_n \rangle$ and a value v

Output: An index i s.t. $v = A[i]$ or the special value NIL if v does not appear in A .

Linear_Search (A, v)

```
i ← NIL
for j = 1 to A.length do
    if A[j] = v then
        i ← j
        return i
    end if
end for
return i
```

Complexity : $O(n)$

Selection Sort

Sort n numbers by iteratively selecting the minimum in subarray & swapping with the first element of subarray

example: In the first iteration: find minimum of $A[1 \dots n]$
& swap with $A[1]$
" " k^{th} " : find minimum of $A[k \dots n]$
& swap with $A[k]$

Selection Sort (A)

```
for  $i = 1$  to  $n-1$  do
     $\text{min} \leftarrow i$ 
    for  $j = i+1$  to  $n$  do
        // find the index of the  $i^{\text{th}}$  smallest element
        if  $A[j] < A[\text{min}]$  then
             $\text{min} = j$ 
        end if
    end for
    Swap  $A[\text{min}]$  with  $A[i]$ 
end for
```

Loop invariant: At each iteration i of the algorithm the subarray $A[1 \dots i-1]$ contains $(i-1)$ smallest elements of A in increasing order.

Running time:

Worst case: $\Theta(n^2)$

Best case: $\Theta(n^2)$

$$T(n) = \sum_{i=1}^{n-1} (n-i)a + b$$

$$\begin{aligned} T(n) &= \sum_{i=1}^{n-1} (n-i) = n(n-1) - \sum_{i=1}^{n-1} i \\ &= (n^2 - n) - \frac{n(n-1)}{2} \\ &= (n^2 - n)/2 = \Theta(n^2) \end{aligned}$$

comparisons in i^{th} iter

Bubble sort

It works by repeatedly swapping adjacent elements that are out of order

Bubblesort (A)

```
for i = 1 to A.length - 1
  for j = A.length downto (i+1)
    if A[j] < A[j-1]
      swap A[j] with A[j-1]
    end if
  end for
end for
```

Loop invariant: At the start of each iteration (outermost)
the subarray $A[1 \dots (i-1)]$ contains $(i-1)$ smallest elements
in sorted order.

Complexity

Worst case = best case

$$T(n) = \Theta(n^2)$$

Fibonacci Numbers

$$\text{Golden ratio } \phi = \frac{(1+\sqrt{5})}{2}$$

Definition: Given two number a, b such that $a > b > 0$
then a & b are in golden ratio if :

$$\frac{a+b}{a} = \frac{a}{b} = \phi$$

→ solve: $1 + \frac{b}{a} = \frac{a}{b} \Rightarrow 1 + \frac{1}{\phi} = \phi \Rightarrow \phi + 1 = \phi^2$
 $\Rightarrow \phi^2 - \phi - 1 = 0$
 $\phi = \left(\frac{1 \pm \sqrt{5}}{2} \right)$ ←

→ A golden ratio can be written as a continued fraction.

$$\phi = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \dots \infty}}}$$

$$\begin{bmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^n = \begin{cases} M^{n/2} \cdot M^{n/2} & \text{if } n \text{ even} \\ M^{(n-1)/2} \cdot M^{(n-1)/2} \cdot M & \text{if } n \text{ odd} \end{cases}$$

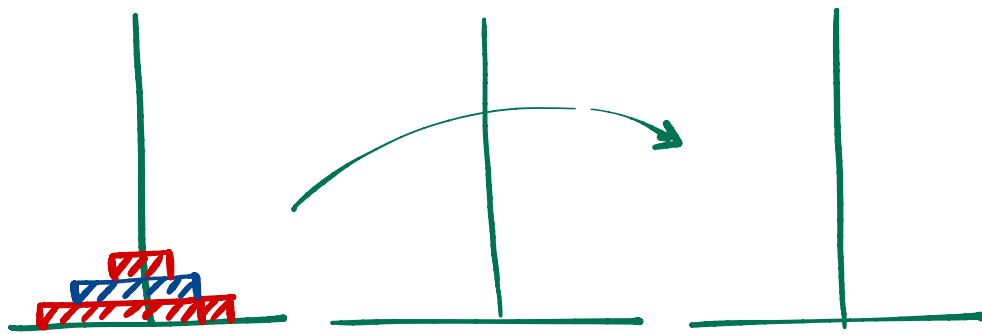
$\downarrow$
 M^n
 $M \in \mathbb{R}^{2 \times 2}$

$\theta(\log n)$

Tower of Hanoi

A puzzle containing 3 rods and a number of discs of various diameters that can slide onto any rod.

- Puzzle begins with all discs stacked on one rod in order of decreasing size; smallest on top.

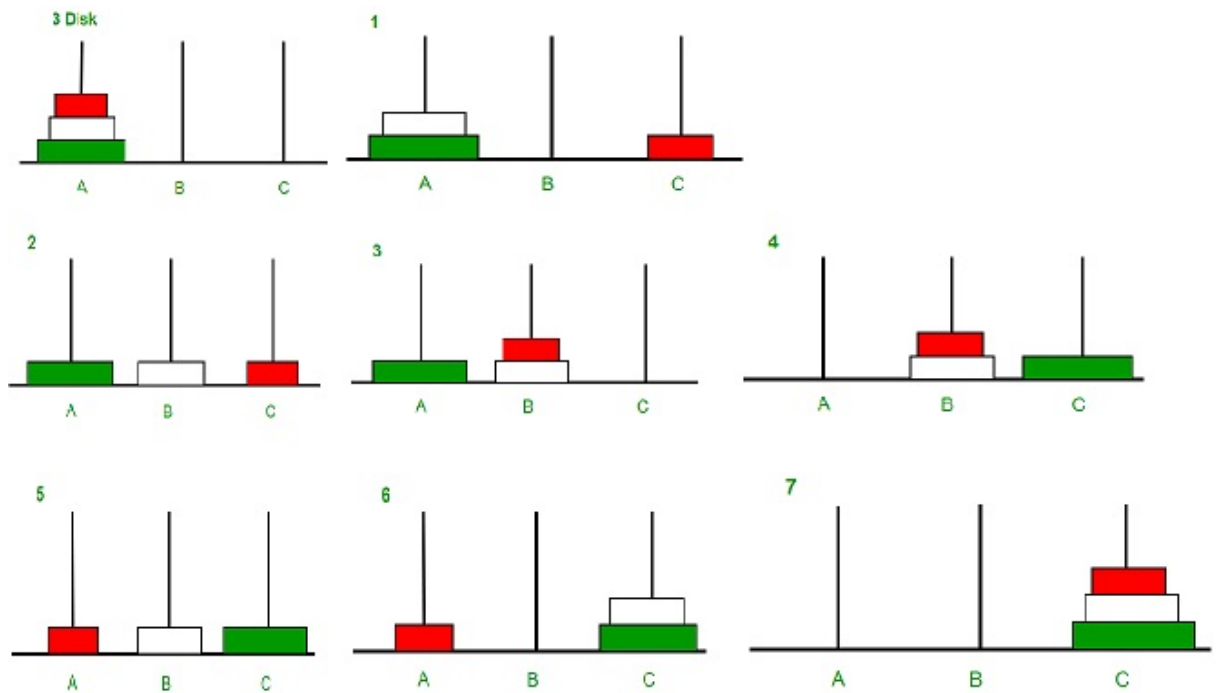


Objective : To move the entire stack to one of the other rods

Rules :

1. Only one disc may be moved at a time
2. Discs can only be moved from the top of a stack to the top of another stack or on an empty stack
3. Only a smaller disc may be placed on top of an existing disc.

— solution for $n=3$:



Complexity : $T(n) = 2^n - 1$ $\rightarrow$ exponential !!

# discs	# steps
$n = 1$	1
$n = 2$	3
$n = 3$	7
$\vdots$	$\vdots$
$n = k$	$2^k - 1$

$\Rightarrow$ In general, we can write the recurrence :

$$\underline{T(n) = 2T(n-1) + 1}$$